

Cool Flames Investigation (CFI) Operational Requirements and Protocols

Operational Requirements:

Droplets must be deployed free floating without any residual momentum in the center of the field of view of the CIR diagnostics. Diagnostics must be capable of recording the radiant emissions of both hot and cool flames. CIR must store be able to store digital imagery data from a full day of test point (approx. 8 burns) for later downlink.

Operational Protocols:

Following setup and initialization of the CIR rack and MDCA, the chamber is filled with the appropriate atmosphere for the day's test points. Conditions vary in pressure (0.5 atm to 5.0 atm) and oxygen concentration (10 to 30%). A settling time of approximately 2 minutes elapses prior to initiating the test in order to ensure that the temperature and pressure of the chamber gases have stabilized. This settling time is followed by the dispensing of a predetermined amount of fuel (based on the target droplet size) onto a support fiber. When sufficient fuel has been dispensed the dispensing needles are retracted and a dwell period of at least 10 seconds is allowed for the droplet internal fluid motion induced by deployment to subside. This is then be followed by initiating power to an igniter pair for a selectable amount of time ranging from 1 second to 5 seconds after which the igniter will be retracted from the field of view. A near real-time download of the color camera video is required in order to verify successful droplet deployment, ignition, and overall progress of the experiment. Pressure and temperature data of the chamber environment is also required in near real time.